1. The Poisson Distribution

1.1. Introduction

- The **Poisson distribution** is used as a model for the number , X, of events in a given interval of space or times. It has the probability formula

$$P(X=x)=e^{-\lambda}\frac{\lambda^x}{x!} \qquad x=0,\ 1,\ 2,\ldots$$

Where λ is equal to the mean number of events in the given interval

- A Poisson distribution with mean λ can be noted as

$$X\ \sim\ Po(\lambda)$$

1.2. Suitability of a Poisson Distribution

- Occur randomly in space or time
- Occur singly – events cannot occur simultaneously
- Occur independently
- Occur at a constant rate – mean no. of events in given time interval proportional to size of interval

1.3. Expectation & Variance

- For a Poisson distribution $X \sim Po\ (\lambda)$
- Mean $=\mu=E(X)=\lambda$
- Variance $=\sigma^2=Var(X)=\lambda$
- The mean & variance of a Poisson distribution are equal

1.4. Addition of Poisson Distributions

- If X and Y are independent Poisson random variables, with parameters λ and μ respectively, then $X+Y$ has a Poisson distribution with parameter $\lambda+\mu$

{IS Ex 8d} Question 1:

The numbers of emissions per minute from two radioactive objects A and B are independent Poisson variables with mean 0.65 and 0.45 respectively. Find the probabilities that:

1. In a period of three minutes there are at least three emissions from A.
2. In a period of two minutes there is a total of less than four emissions from A and B together.

Solution:

Part (i):

Write the distribution using the correct notation

$$A\sim Po(0.65\times 3)=\ A\sim Po(1.95)$$

Use the limits given in the question to find probability

$$P(A\geq 3)=1-P(A<3)$$

$$=1-\left(\frac{1.95^2e^{-1.95}}{2!}+\frac{1.95^1e^{-1.95}}{1!}+\frac{1.95^0e^{-1.95}}{0!}\right)$$

$$=1-0.690=0.310$$

Part (ii):

Write the distribution using the correct notation

$$(A+B)\sim Po(2(0.65+0.45))=(A+B)\sim Po(2.2)$$

Use the limits given in the question to find probability

$$P(A<4)=e^{-2.2}\left(\frac{(2.2)^3}{3!}+\frac{(2.2)^2}{2!}+\frac{(2.2)^1}{1!}+\frac{(2.2)^0}{0!}\right)$$

$$=0.819$$

1.5. Relationship of Inequalities

- $P(X<r)=P(X\leq r-1)$
- $P(X=r)=P(X\leq r)-P(X\leq r-1)$
- $P(X>r)=1-P(X\leq r)$
- $P(X\geq r)=1-P(X\leq r-1)$

1.6. Poisson Approximation of a Binomial Distribution

- To approximate a binomial distribution given by:

$$X\sim B(n,p)$$

- If $n>50$ and $np<5$
- Then we can use a Poisson distribution given by:

$$X\sim Po(np)$$

{IS Ex 8d} Question 8:

A randomly chosen doctor in general practice sees, on average, one case of a broken nose per year and each case is independent of the other similar cases.

1. Regarding a month as a twelfth part of a year,
    1. Show that the probability that, between them, three such doctors see no cases of a broken nose in a period of one month is 0.779
    2. Find the variance of the number of cases seen by three such doctors in a period of six months
2. Find the probability that, between them, three such doctors see at least three cases in one year.
3. Find the probability that, of three such doctors, one sees three cases and the other two see no cases in one year.

Solution:

Part (i)(a):

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Write down the information we know and need

1 doctor $= 1$ nose per year $= \frac{1}{12}$ noses per month

3 doctors $= \frac{3}{12} = \frac{1}{4}$ noses per month

Write the distribution using the correct notation

$$X \sim Po\,(0.25)$$

Use the limits given in the question to find probability

$$P\,(X = 0) = \frac{0.25^0 e^{-0.25}}{0!} = 0.779$$

Part (i)(b):

Use the rules of a Poisson distribution

$$\text{Var}\,(X) = \mu = \lambda$$

Calculate λ in this scenario:

$$\lambda = 6 \times \mu \text{ (in one month)} = 6 \times 0.25 = 1.5$$

$$\therefore Var\,(X) = 1.5$$

Part (ii):

Calculate λ in this scenario:

$$\lambda = 12 \times \mu \text{ (in one month)} = 12 \times 0.25 = 3$$

Use the limits given in the question to find probability

$$P\,(X \geq 3) = 1 - P\,(X \leq 2)$$

$$= 1 - e^{-3}\left(\frac{3^2}{2!} + \frac{3^1}{1!} + \frac{3^0}{0!}\right) = 1 - 0.423 = 0.577$$

Part (iii):

$$\lambda \; for \; one \; doctor \; in \; one \; year = 1$$

$$\lambda \; for \; other \; two \; doctors \; in \; one \; year = 2 \times 1 = 2$$

For the first doctor:

$$P\,(X = 3) = e^{-1}\left(\frac{1^3}{3!}\right)$$

For the two other doctors:

$$P\,(X = 0) = e^{-1}\left(\frac{1^0}{0!}\right)$$

Considering that any of the three could be the first

$$P\,(X) = e^{-1}\left(\frac{1^3}{3!}\right) \times e^{-1}\left(\frac{1^0}{0!}\right) \times {}^3C_2 = 0.025$$

1.7. Normal Approximation of a Poisson Distribution

- To approximate a Poisson distribution given by:

$$X \sim P(\lambda)$$

- If $\lambda > 15$
- Then we can use a normal distribution given by:

$$X \sim N(\lambda, \lambda)$$

Apply continuity correction to limits:

Poisson	Normal
$x = 6$	$5.5 \leq x \leq 6.5$
$x > 6$	$x \geq 6.5$
$x \geq 6$	$x \geq 5.5$
$x < 6$	$x \leq 5.5$
$x \leq 6$	$x \leq 6.5$

{IS Ex 10h} Question 11:

The no. of flaws in a length of cloth, lm long has a Poisson distribution with mean $0.04l$

1. Find the probability that a 10m length of cloth has fewer than 2 flaws.
2. Find an approximate value for the probability that a 1000m length of cloth has at least 46 flaws.
3. Given that the cost of rectifying x flaws in a 1000m length of cloth is x^2 pence, find the expected cost.

Solution:

Part (i):

Form the parameters of Poisson distribution

$l = 10$ and $\lambda = 0.04l$

$$\therefore \lambda = 0.4$$

Write down our distribution using correct notation

$$X \sim Po\,(0.4)$$

Write the probability required by the question

$$P\,(X < 2)$$

From earlier equations:

$$P\,(X < 2) = e^{-0.4}\left(\frac{0.4^0}{0!} + \frac{0.4^1}{1!}\right) = 0.938$$

Part (ii):

Using question to form the parameters

$l = 1000$ and $\lambda = 0.04l$

$$\therefore \lambda = 40 > 15$$

Thus, we can use the normal approximation

Write down our distribution using correct notation

$$X \sim Po\,(40) \rightarrow Y \sim N\,(40,\ 40)$$

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Write the probability required by the question

$$P(X \geq 46)$$

Apply continuity correction for the normal distribution

$$P(Y \geq 45.5)$$

Evaluate the probability

$$P(Y \geq 45.5) = 1 - \Phi\left(\frac{45.5 - 40}{\sqrt{40}}\right) = 0.192$$

Part (iii):

Using the variance formula

$$\text{Var}(X) = E\left(X^2\right) - (E(X))^2$$

For a Poisson distribution

$E(X) = Var(X) = \lambda$ and $\lambda = 40$

Substitute into equation and solve for the unknown

$$\therefore 40 = E\left(X^2\right) - 40^2$$

$E\left(X^2\right) = 1640$ pence

$$E\left(X^2\right) = £16.40$$

Expected cost for rectifying cloth is $£16.40$

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